

FINAL PROJECT REPORT

SMARTPHONE USAGE, DIGITAL ADDITCTION & WELL-BEING ANALYTICS

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1. Introduction

The Smartphone Usage, Digital Addiction & Well-Being Analytics project focuses on understanding how people use smartphones in their daily lives and how smartphone usage may be related to different aspects of their well-being. Smartphones have become an important part of everyday life. People use them for communication, education, entertainment, social media, online shopping, banking, work, navigation, and many other activities. Because smartphones are used for many hours every day, it is important to understand usage patterns and identify whether excessive usage may be associated with negative effects on daily life.

This project uses data analytics to study smartphone usage patterns from a given dataset. The dataset contains information related to smartphone usage and different well-being factors. By analyzing this data, meaningful patterns and relationships can be identified. Instead of looking at individual records manually, data analytics helps organize, process, compare, and interpret large amounts of information in a systematic way.

The main idea of the project is to examine how much people use their smartphones, what activities they perform on their smartphones, and how these usage patterns are connected with their digital well-being. Different users may have completely different smartphone habits. Some users may mainly use smartphones for education or work, while others may spend more time on social media, entertainment, gaming, or other activities. Therefore, analyzing usage data can help provide a clearer understanding of these different patterns.

The project also focuses on the concept of digital addiction. Digital addiction refers to excessive or difficult-to-control use of digital devices or digital services that may interfere with normal daily activities. Smartphone usage by itself does not necessarily mean addiction. However, very high usage, continuous engagement, difficulty reducing usage, or smartphone use interfering with other areas of life can be important patterns to examine. Through analytics, the project attempts to identify usage patterns that may indicate higher levels of digital dependence.

Another important part of the project is well-being. A person's digital habits can be studied together with available well-being-related information in the dataset. This makes it possible to explore whether differences in smartphone usage are associated with differences in factors such as sleep, productivity, mood, or other available well-being measures. The project does not simply focus on the amount of smartphone usage; instead, it considers multiple factors together to obtain a broader understanding of digital behavior.

The project follows a data-driven approach. First, the available dataset is examined to understand the different columns, variables, and records. The data is then cleaned and prepared so that it can be analyzed properly. After preprocessing, different analytical techniques can be applied to identify patterns, relationships, and trends within the data.

SQL is an important part of this project because it can be used to store, organize, retrieve, filter, and analyze structured data. SQL queries can be used to answer questions such as which users have higher usage levels, which activities consume more time, how usage differs between groups, and how different variables are related. This makes SQL useful for extracting meaningful information from the dataset.

Data visualization can also be used to present the results in an understandable manner. Tables, charts, graphs, and dashboards can make complex numerical information easier to interpret. For example, a chart can be used to compare average smartphone usage across different categories, while another visualization can show the relationship between smartphone usage and a well-being-related factor.

The project also provides an opportunity to understand the complete data analytics workflow. It involves collecting or receiving data, understanding the dataset, cleaning the data, transforming the data when required, storing or managing the data, performing analysis, identifying useful insights, and presenting the final results. Therefore, the project is not limited to creating charts; it demonstrates how raw data can be converted into meaningful information.

The results obtained from this analysis can help users and researchers better understand smartphone usage behavior. The findings may also help identify patterns that deserve further attention, particularly when smartphone usage appears to be

associated with reduced productivity, disturbed routines, or other well-being-related factors present in the dataset.

Overall, the Smartphone Usage, Digital Addiction & Well-Being Analytics project aims to provide a structured and data-based understanding of smartphone usage and its relationship with digital well-being. By combining data analysis, SQL, data visualization, and meaningful interpretation, the project transforms raw smartphone usage data into useful insights. It demonstrates how analytics can be applied to a real-world social and technological problem and how data can support a better understanding of people's digital habits.

In simple words

This project answers questions such as:

- How much are people using their smartphones?
- What are they mainly using their smartphones for?
- Which users show higher levels of smartphone usage?
- Are there noticeable patterns in excessive smartphone use?
- How does smartphone usage relate to the well-being information available in the dataset?
- What useful patterns can be discovered from the data?
- How can SQL and data visualization help explain these patterns?

1.1 Project Overview

The Smartphone Usage, Digital Addiction & Well-Being Analytics project is a data analytics-based project designed to study and understand patterns in smartphone usage and their relationship with digital well-being. Smartphones are now widely used for communication, education, entertainment, social networking, online services, and many other daily activities. As the amount of time spent on smartphones continues to increase, analyzing smartphone usage data can provide useful information about people's digital habits and behavior.

The project uses a structured dataset containing information related to smartphone usage and well-being. The available data is examined carefully to understand the different attributes, values, and relationships present in the dataset. Each record represents information that can be analyzed to identify patterns in smartphone usage and other related factors. The project converts this raw data into meaningful information through a systematic data analytics process.

The first stage of the project involves understanding and preparing the dataset. Raw data may contain missing values, inconsistent values, duplicate records, or other issues that can affect the accuracy of analysis. Therefore, the dataset is examined and cleaned before performing detailed analysis. Data preprocessing helps ensure that the information used for analysis is organized and suitable for generating meaningful results.

After preparing the data, different data analysis techniques are used to study smartphone usage patterns. The analysis can include examining usage levels, comparing different categories of users, identifying frequently occurring patterns, and studying relationships between smartphone-related variables and well-being-related variables available in the dataset. These analyses help in understanding how smartphone behavior differs among users.

A major focus of the project is the analysis of digital addiction-related patterns. The project examines smartphone usage behavior to identify patterns associated with excessive or intensive usage. Instead of considering only one factor, multiple available variables can be examined together to obtain a broader view of digital behavior. This allows the project to investigate whether certain usage patterns are associated with other changes in the well-being-related information contained in the dataset.

The project also considers digital well-being as an important area of analysis. Smartphone usage can be studied alongside available well-being indicators to understand possible relationships within the data. For example, if the dataset contains information related to sleep, productivity, mood, or other well-being measures, these variables can be compared with smartphone usage patterns. Such analysis can help identify trends and relationships present in the dataset without relying only on assumptions.

SQL is used as an important analytical technology in the project. The dataset can be organized into a structured database, allowing SQL queries to be used for retrieving and analyzing information. SQL can help perform operations such as filtering records, grouping data, calculating averages, counting records, sorting information, and comparing different categories. These operations make it easier to extract useful information from the dataset.

The project also uses data visualization to represent analytical results in a simple and understandable form. Numerical results obtained from the analysis can sometimes be difficult to understand when presented only as tables. Charts and graphs can make comparisons, trends, and relationships easier to identify. Appropriate visualizations can therefore be created to present important findings from the smartphone usage data.

Another important aspect of the project is the creation of meaningful analytical insights. The purpose is not simply to calculate values or create graphs, but to understand what the analyzed data indicates. For example, the analysis may reveal differences in smartphone usage between groups, identify activities associated with higher usage, or show relationships between usage patterns and available well-being indicators. These findings can then be explained in a clear and understandable manner.

The project follows a complete data analytics workflow, beginning with the available dataset and continuing through data understanding, data cleaning, data preparation, SQL-based analysis, visualization, interpretation, and presentation of findings. Each stage contributes to converting raw information into useful insights.

The project can also provide a practical understanding of how data analytics can be applied to a real-world technology-related problem. Instead of working only with theoretical examples, the project deals with smartphone usage data and attempts to discover meaningful patterns from it. This demonstrates the practical application of database management, SQL, data analysis, and visualization techniques.

The final outcome of the project is expected to be a collection of organized analysis, visual representations, and meaningful findings related to smartphone usage, digital addiction-related patterns, and well-being. The results can help provide a clearer picture of digital usage behavior and can serve as a basis for understanding the patterns present in the given dataset.

In summary, the Smartphone Usage, Digital Addiction & Well-Being Analytics project provides a structured approach to analyzing smartphone-related data. It combines data preprocessing, SQL, analytical techniques, and visualization to study smartphone usage patterns and their relationship with available well-being information. By transforming raw dataset records into understandable insights, the project demonstrates how data analytics can be used to study digital behavior and well-being in a systematic and data-driven manner.

1.2 Objectives

The main objective of the Smartphone Usage, Digital Addiction & Well-Being Analytics project is to analyze smartphone usage data and identify meaningful patterns related to digital usage behavior and well-being. The project aims to transform the available raw dataset into useful and understandable information by applying data analytics and SQL techniques.

The major objectives of the project are as follows:

1. To understand smartphone usage patterns

The project aims to examine how smartphones are being used by different users. The available data is analyzed to understand usage levels, activities, and other smartphone-related factors present in the dataset. Studying these patterns helps provide a clear picture of general smartphone usage behavior.

2. To analyze excessive smartphone usage

The project aims to identify patterns associated with high or excessive smartphone usage. By examining the available usage-related attributes, the project attempts to understand which users or categories show higher levels of smartphone

engagement. This helps in studying digital usage behavior from a data-driven perspective.

3. To study digital addiction-related patterns

Another important objective is to analyze patterns that may be associated with digital addiction or excessive dependence on smartphones. Different variables in the dataset can be examined together rather than depending on a single measure. This allows the project to identify combinations of usage behaviours that may require further attention.

4. To examine the relationship between smartphone usage and well-being

The project aims to study the relationship between smartphone usage and the well-being-related information available in the dataset. Variables related to well-being can be compared with smartphone usage measures to identify noticeable patterns or relationships within the data.

5. To clean and prepare the dataset

An important objective of the project is to make the available dataset suitable for analysis. The data is examined for issues such as missing values, duplicate records, inconsistent entries, or inappropriate formats where applicable. Proper data preparation helps improve the reliability and usefulness of the subsequent analysis.

6. To use SQL for data analysis

The project aims to apply SQL for storing, retrieving, filtering, grouping, and analyzing the structured data. SQL queries can be used to obtain specific information from the dataset and perform calculations and comparisons. This demonstrates the practical use of SQL in a real-world data analytics project.

7. To identify important trends and relationships

The project aims to discover useful trends, patterns, and relationships within the dataset. Instead of simply displaying individual records, the analysis focuses on finding information that can provide a better understanding of smartphone usage and well-being.

8. To compare different categories of users

The project aims to compare users or groups based on the categories and variables available in the dataset. Such comparisons can help identify differences in smartphone usage behavior and other related factors. This can provide a clearer understanding of how usage patterns vary among different groups.

9. To present data through visualizations

Another objective is to represent important analytical results using suitable charts, graphs, and other visual formats. Visualization makes numerical information easier to understand and helps users identify differences, trends, and relationships more quickly.

10. To generate meaningful insights from the data

The project aims to convert raw data into meaningful insights. The results of the analysis should not be limited to numerical calculations; they should also explain what the observed patterns indicate about smartphone usage, digital addiction-related behavior, and well-being factors present in the dataset.

11. To provide a data-driven understanding of digital well-being

The project aims to provide an overall understanding of digital well-being based on the information available in the dataset. By bringing together smartphone usage and well-being-related variables, the project can provide a broader view of digital behavior and its observed patterns.

12. To demonstrate the practical application of data analytics

The project also aims to demonstrate how data analytics can be applied to a real-world problem. It combines data preparation, SQL-based analysis, data interpretation, and visualization to show how raw data can be transformed into useful information.

Overall Objective

The overall objective of the project is to analyze smartphone usage data, identify digital addiction-related patterns, examine available well-being factors, and generate meaningful insights using data analytics and SQL. The project provides a systematic and data-driven approach to understanding smartphone behavior and the patterns associated with digital well-being.

2. Project Initialization and Planning Phase

The Project Initialization and Planning Phase is the first major phase of the project development process. It establishes a clear foundation for the Smartphone Usage, Digital Addiction & Well-Being Analytics project. During this phase, the project idea is understood, the problem to be addressed is identified, the objectives are defined, the available data is examined at a basic level, and the overall approach for completing the project is planned.

Proper planning is important because a data analytics project involves several stages, such as understanding the data, preparing the data, performing analysis,

writing SQL queries, creating visualizations, interpreting the results, and documenting the findings. If these activities are not planned properly, the project may become difficult to manage and the final analysis may not produce meaningful results.

The initialization phase begins with identifying the main area of the project. In this project, the selected area is smartphone usage, digital addiction, and well-being. The project focuses on using available data to understand smartphone-related behavior and to examine patterns associated with digital well-being.

The project is then defined in terms of its purpose and expected outcome. The purpose is not simply to store or display smartphone usage data. The project aims to analyze the available information and convert it into meaningful insights. The expected outcome includes organized data, analytical results, suitable visualizations, and understandable findings related to smartphone usage and the well-being-related variables available in the dataset.

Identification of the Problem

The first important activity in the planning phase is identifying the problem that the project is intended to address.

Smartphones are used for many different purposes, including communication, education, entertainment, social networking, work, and online services. Because of this wide range of uses, smartphone usage patterns can vary significantly between users. Some users may spend a limited amount of time on their devices, while others may have much higher levels of usage.

Simply knowing that a smartphone is being used frequently is not enough to understand the complete situation. Different factors need to be examined together. Therefore, the project focuses on analyzing the available data to identify usage patterns and examine how those patterns are related to the well-being information contained in the dataset.

The problem is therefore approached as a data analytics problem, where available smartphone-related data is studied systematically to discover useful patterns and relationships.

Defining the Project Scope

The project scope defines what the project will cover and what will be considered during the analysis.

The scope of this project includes the analysis of the provided smartphone usage dataset. The available attributes are examined to understand smartphone usage behavior and relevant well-being-related information. Data cleaning and preparation are performed where required before analysis.

The project includes the use of SQL and data analytics techniques for extracting and analyzing information from the dataset. Appropriate visualizations can also be created to communicate important findings clearly.

The project focuses on the information available in the dataset. Therefore, conclusions are based on the data that is actually provided rather than assuming information that is not present.

Understanding the Available Dataset

After defining the project scope, the available dataset is examined to understand its structure.

The dataset is reviewed to identify the different columns, types of information, number of records, and values stored in each field. This initial examination helps in understanding what type of analysis can be performed.

The dataset may contain different categories of information related to smartphone usage and well-being. Each variable is considered according to its meaning and relevance to the project. Understanding the meaning of each field is important because incorrect interpretation of a variable can lead to incorrect analysis.

At this stage, the data is mainly studied to understand its structure. Detailed cleaning and analysis are performed in later stages of the project.

Defining Project Objectives

Once the problem and dataset are understood, the objectives of the project are established.

The objectives provide clear direction for the project and help determine what needs to be achieved. The objectives include understanding smartphone usage patterns, analyzing excessive usage-related patterns, examining digital addiction-related behavior, studying available well-being factors, using SQL for data analysis, identifying meaningful relationships, and presenting the results through suitable visualizations.

Clearly defined objectives also help in evaluating whether the project has successfully achieved its intended purpose.

Planning the Data Analytics Process

The project is planned as a sequence of connected activities. The general workflow includes:

Dataset → Data Understanding → Data Cleaning → Data Preparation → SQL/Data Analysis → Visualization → Interpretation → Final Insights → Documentation

Each stage has a specific purpose.

First, the dataset is understood. Next, data quality problems are identified and addressed. After the data is prepared, analytical operations are performed. SQL can be used to retrieve and analyze relevant information. The results can then be represented using charts and graphs. Finally, the results are interpreted and documented.

Following a structured workflow helps ensure that important steps are not missed.

Selection of Tools and Technologies

The planning phase also involves identifying the technologies required for the project.

Excel can be used initially to examine and understand the provided dataset. It can help in viewing records, identifying columns, checking values, and performing basic data inspection.

SQL is used for structured data analysis. It allows the project to perform operations such as filtering records, grouping information, calculating values, sorting results, and retrieving specific information from the database.

Data visualization tools or suitable visualization techniques can be used to present important findings in graphical form. The exact visualization method can be selected according to the type of data and the analytical question being addressed.

GitHub can be used for maintaining and organizing the project work, including relevant code, queries, documentation, and project files where appropriate.

Planning the Data Cleaning Process

Data quality is an important consideration during project planning. Before performing analysis, the dataset needs to be checked for possible problems.

The planning process considers issues such as:

- Missing values
- Duplicate records
- Incorrect or inconsistent entries
- Different formats for similar values
- Unnecessary columns
- Incorrect data types
- Invalid or unexpected values

Not every dataset will contain all these problems. Therefore, the actual cleaning steps should be performed according to the problems found in the provided dataset.

The purpose of data cleaning is to make the dataset more consistent and suitable for analysis.

Planning the SQL Analysis

SQL analysis is planned according to the questions that the project needs to answer.

Examples of analytical questions may include identifying usage patterns, calculating average values, comparing different groups, counting records according to categories, and examining relationships between relevant variables.

SQL queries can be developed progressively. Basic queries can first be used to understand the data, followed by filtering, grouping, aggregation, sorting, and more advanced queries where required.

This approach ensures that SQL is used not just as a technical requirement but as a practical tool for extracting meaningful information from the dataset.

Planning Data Visualization

The planning phase also determines how analytical results will be presented.

Different types of information may require different types of visualizations. For example, bar charts can be useful for comparing categories, while other charts may be suitable for showing distributions or relationships between variables.

The objective of visualization is to make the results easier to understand. A well-designed visualization can help identify patterns that may not be immediately visible when looking at raw data or large tables.

Only relevant visualizations should be included so that the final presentation remains clear and meaningful.

Planning the Project Schedule

The project is divided into smaller activities so that the work can be completed systematically.

A general project schedule can include:

1. Understanding the project topic
2. Defining the problem and objectives
3. Examining the dataset
4. Understanding the variables

5. Cleaning and preparing the data
6. Organizing the data for analysis
7. Performing SQL analysis
8. Identifying important findings
9. Creating visualizations
10. Interpreting the results
11. Preparing the final documentation
12. Reviewing and presenting the completed project

Breaking the project into these stages makes the overall work easier to manage and monitor.

Expected Outcomes

During the planning phase, the expected outcomes of the project are also identified.

The project is expected to produce a structured analysis of smartphone usage and the relevant well-being information contained in the dataset. It should identify useful patterns and relationships and present them in an understandable form.

The final project is also expected to demonstrate the practical application of data analytics and SQL. The analysis should show how raw data can be transformed into meaningful information through proper data preparation, analysis, visualization, and interpretation.

Importance of the Planning Phase

The Project Initialization and Planning Phase provides a clear direction for the entire project. It prevents the project from becoming a collection of unrelated analyses and ensures that each activity is connected to the defined objectives.

A proper plan also helps in selecting appropriate tools, organizing the dataset, deciding what analysis should be performed, and determining how the final results should be presented.

For the Smartphone Usage, Digital Addiction & Well-Being Analytics project, this phase establishes the foundation for studying smartphone usage behavior using a systematic data-driven approach. The decisions made during initialization and planning guide the subsequent phases of data preparation, analysis, visualization, and documentation.

2.1 Define Problem Statement

The Smartphone Addiction & Digital Well-Being Insights project is developed to understand and analyze the increasing use of smartphones and the different patterns associated with users' digital activities. Smartphones have become an important part of everyday life and are used for communication, social networking, entertainment, gaming, education, work, and many other activities. Because of this wide usage, it is important to understand how users spend their time on smartphones and whether their usage patterns indicate excessive or potentially addictive behavior.

The major problem addressed by this project is that raw smartphone usage data can contain a large number of different measurements, but it is difficult to understand the overall behavior of users by looking at individual values or records. Information such as daily screen time, weekend screen time, social media hours, gaming hours, sleep hours, notifications, app openings, stress level, age, gender, and addiction level needs to be analyzed together to obtain meaningful insights.

Excessive smartphone usage may be associated with spending more time on digital activities and less time on other daily activities. Therefore, the project focuses on studying different aspects of smartphone usage rather than considering screen time alone. By examining screen time together with social media usage, gaming, sleep, notifications, app openings, and other available measures, the project provides a broader understanding of digital usage behavior.

Another important problem is the difficulty of identifying patterns among different categories of users. Users may have different ages, genders, levels of smartphone usage, and addiction levels. A simple overall average may not provide enough information to understand these differences. Therefore, the project uses visual analysis to compare usage patterns between different user groups and categories.

The project also addresses the need to understand the relationship between smartphone usage and digital well-being. Digital well-being refers to maintaining a balanced and healthy relationship with digital devices and technology. In this project, digital well-being is studied through measurable factors available in the dataset, such as screen time, sleep hours, social media usage, gaming hours, notifications, app openings, stress levels, and addiction-related information.

Detailed Problem Description

The problem can be understood through the following points:

1. Difficulty in understanding smartphone usage

Users may use smartphones for several different purposes throughout the day. Without proper analysis, it is difficult to determine how much time is spent on different activities and which activities contribute more to overall smartphone usage.

2. Excessive screen-time patterns

Daily screen time and weekend screen time are important indicators of smartphone usage. Analyzing these values helps identify general usage patterns and understand how smartphone usage changes during different periods.

3. Multiple digital activities

Smartphone usage is not limited to a single activity. Users may spend time on social media, gaming, entertainment, communication, education, or other applications. Therefore, the problem requires analysis of multiple activities instead of focusing only on total screen time.

4. Understanding addiction levels

The dataset contains information related to addiction levels and addicted status. However, these categories can be difficult to understand when presented as raw records. Visualization can make it easier to observe how users are distributed across different addiction categories.

5. Comparing addicted and non-addicted users

Another part of the problem is understanding the difference between users who are categorized as addicted and those who are not. A visual representation can make the proportion and distribution of these categories easier to understand.

6. Age-wise differences

Smartphone usage may vary across different age groups. Therefore, the project analyzes age groups and their corresponding screen-time patterns to understand how usage is distributed among different groups.

7. Gender-wise analysis

The project also considers gender as one of the available categories. Comparing average usage values across gender categories can provide a clearer view of the distribution of smartphone usage within the dataset.

8. Social media usage

Social media is an important part of smartphone activity. The project analyzes social media usage hours to understand its contribution to users' overall digital activity.

9. Gaming behavior

Gaming hours are also considered as an important usage measurement. Analyzing gaming together with other activities helps provide a more complete picture of smartphone behavior.

10. Sleep-related analysis

Sleep hours are included as another factor in the analysis. The project examines screen time and sleep-related data to identify patterns present within the dataset.

11. Notifications and application openings

The number of notifications received and the number of applications opened provide additional information about how frequently users interact with their smartphones. These measurements are included to understand the intensity of digital interaction.

12. Stress-related information

Where available in the dataset, stress level is considered as another factor associated with users' digital behavior. It can be analyzed alongside other usage measurements to identify patterns in the available data.

13. Difficulty in analyzing raw data

Large amounts of raw data are difficult to understand quickly. Looking at rows and columns of numerical values does not immediately provide a clear picture of the major trends. Therefore, the data needs to be transformed into meaningful charts, graphs, and summary indicators.

14. Lack of centralized visualization

When different measurements are viewed separately, it becomes difficult to compare them. A centralized Power BI dashboard allows important information to be displayed together in one place.

15. Need for interactive analysis

Different users may want to examine different aspects of the dataset. For example, analysis may need to be viewed according to gender, age group, addiction level, or other available categories. Interactive filters and slicers make this type of exploration easier.

16. Difficulty in identifying important patterns

Important information can be hidden inside large datasets. Visualization techniques such as bar charts, column charts, donut charts, scatter charts, KPI cards, and other dashboard elements help make patterns more visible.

17. Need for meaningful business questions

The project does not simply display data. It converts the problem into specific questions, such as:

- What is the average daily screen time?
- How does screen time differ between gender categories?
- What are the different addiction levels?

- What proportion of users are categorized as addicted and non-addicted?
- How does screen time vary across age groups?
- What relationship can be observed between screen time and sleep hours?
- How do gaming hours compare with social media hours?
- What is the average number of notifications received?
- What are the overall digital usage patterns?

18. Need for an easy-to-understand reporting system

The final analysis should be understandable not only to technical users but also to people who want to quickly understand the main findings. Therefore, the project presents the analysis through a simple and interactive Power BI dashboard.

Expected Outcome

The problem-solving process is expected to produce:

- A cleaned and organized smartphone usage dataset.
- Analysis of important smartphone usage measurements.
- Comparison of usage patterns across available user categories.
- Analysis of addiction levels and addicted status.
- Analysis of daily and weekend screen time.
- Analysis of social media and gaming usage.
- Analysis of screen time and sleep-related patterns.
- Analysis of notifications and application openings.
- Easy-to-understand charts and graphs.
- KPI cards showing important summary values.
- Interactive filters and dashboard components.
- A centralized Power BI dashboard for exploring digital usage behavior.
- A clear report describing the observations obtained from the available data.

2.2 Project Proposal (Proposed Solution)

The proposed solution for the Smartphone Addiction & Digital Well-Being Insights project is to develop a data analysis and interactive visualization system using Microsoft Power BI. The purpose of the proposed solution is to organize smartphone usage data, analyze important digital usage patterns, and present the results in a simple and understandable dashboard.

The proposed system will take smartphone usage data as input and process the available information related to daily screen time, weekend screen time, social media usage, gaming hours, sleep hours, notifications, app openings, stress levels, age, gender, addiction levels, and addicted status. Instead of analyzing these values

individually from a raw dataset, the proposed solution will combine them into meaningful visualizations.

The solution will follow a systematic process beginning with data collection and data understanding. The collected dataset will be examined to understand its structure, columns, data types, and available information. The data will then be checked for problems such as missing values, duplicate records, inconsistent values, and incorrect data types. After preprocessing, the prepared data will be used for analysis and visualization.

Proposed Solution

The proposed solution consists of the following major stages:

1. Data Collection

The first stage is to collect the required smartphone usage data. The dataset should contain information that helps analyze users' digital behavior. Important data fields used in the project include screen time, social media usage, gaming, sleep, notifications, application openings, addiction-related information, and user categories such as age and gender.

The collected data will act as the main input for the project. Having organized and relevant data is important because the quality of the analysis depends on the quality and completeness of the available information.

2. Data Understanding

After collecting the data, the dataset will be examined to understand what information is available. The different columns and their meanings will be identified, and the numerical and categorical information will be distinguished.

For example, numerical information such as daily screen time, gaming hours, social media hours, sleep hours, notifications, and app openings can be used for calculations and comparisons. Categorical information such as gender, age group, addiction level, and addicted status can be used to divide users into different categories.

3. Data Cleaning and Preprocessing

The raw data may contain missing values, duplicate records, inconsistent entries, or incorrect data types. Therefore, the proposed solution includes a preprocessing stage before creating the final visualizations.

The preprocessing activities include:

- Checking for missing values.
- Identifying duplicate records.
- Checking incorrect or inconsistent values.

- Converting columns into appropriate data types.
- Standardizing values where required.
- Preparing the dataset for analysis.
- Ensuring that the data can be used correctly in Power BI.

This stage helps prepare the dataset for further analysis.

4. Data Analysis

Once the data has been prepared, different aspects of smartphone usage will be analyzed. The analysis will focus on understanding the major measurements available in the dataset.

The proposed analysis includes:

- Average daily screen time.
- Average weekend screen time.
- Social media usage.
- Gaming hours.
- Sleep hours.
- Notifications received per day.
- Application openings.
- Addiction levels.
- Addicted and non-addicted user categories.
- Screen-time patterns across age groups.
- Screen-time patterns across gender categories.
- Available stress-level information.
- Other relevant relationships present in the dataset.

The purpose is to convert individual data records into meaningful summaries and comparisons.

5. Business Questions

The proposed solution will address specific business or analytical questions to make the analysis more meaningful. Examples include:

1. What is the average daily screen time among users?
2. How does average screen time vary across gender categories?
3. What are the different addiction levels among users?
4. What is the proportion of addicted and non-addicted users?
5. How does screen time vary across different age groups?
6. What relationship can be observed between screen time and sleep hours?
7. How do gaming hours compare with social media hours?
8. What is the average number of notifications received per day?
9. What are the overall digital usage patterns of the users?
10. Which available factors show notable patterns in relation to addicted status?

These questions provide a clear direction for developing the dashboard and selecting appropriate visualizations.

6. Data Visualization

The proposed solution will use Microsoft Power BI to convert the analyzed data into visual form. Different visualization types will be selected according to the type of information being presented.

The dashboard can include:

- Column charts for comparing average screen time across categories.
- Bar charts for displaying addiction-level distributions.
- Donut charts for showing proportions such as addicted and non-addicted users.
- Scatter charts for examining relationships such as screen time and sleep hours.
- KPI cards for displaying important summary values.
- Slicers/filters for allowing users to interact with the dashboard.
- Key Influencers for examining factors associated with the selected outcome, where supported by the available data.

The use of different visualizations will make the analysis easier to understand than presenting the same information only in tables.

7. Interactive Dashboard

A major part of the proposed solution is the development of an interactive Power BI dashboard. The dashboard will bring the important analysis into a single interface.

Users can use available filters or slicers to explore the data according to different categories. For example, the dashboard can allow the analysis to be viewed according to gender, age group, addiction level, or other available categories.

The interactive nature of the dashboard allows users to explore the data instead of viewing only fixed results.

8. KPI-Based Summary

The proposed dashboard will use KPI cards to display important numerical information at a glance. These cards provide a quick summary before the user examines the detailed charts.

The dashboard shown for this project includes measures such as:

- Average daily screen time.
- Average social media hours.
- Average gaming hours.
- Average sleep hours.

- Average notifications.
- Average app openings.
- Average weekend screen time.
- Average work/study hours.
- Total number of users.

These values help provide an overall summary of the dataset.

9. Comparative Analysis

The proposed solution will also provide comparison between different categories. Instead of showing only overall averages, the dashboard can be used to examine how values differ across groups.

For example, users can be compared based on:

- Age groups.
- Gender categories.
- Addiction levels.
- Addicted status.
- Different digital activities.

Such comparisons help identify variations within the available dataset.

10. Relationship Analysis

The proposed solution will examine relationships between selected variables where appropriate. For example, daily screen time and sleep hours can be displayed using a scatter chart.

Similarly, gaming and social media usage can be compared to understand the distribution of these digital activities.

Relationship-based visualizations can provide additional information that may not be visible when individual variables are examined separately.

11. Digital Well-Being Analysis

The proposed solution brings different smartphone usage measurements together to provide a broader view of digital well-being.

Instead of considering only screen time, the analysis includes several available indicators such as:

Screen Time → Social Media → Gaming → Sleep → Notifications → App Usage → Stress → Addiction-related Information

This combined approach provides a more comprehensive understanding of digital usage behavior represented in the dataset.

12. Report Generation

After completing the analysis and dashboard development, the results can be documented in a project report. The report will describe the project objective, problem statement, proposed solution, data collection, preprocessing, analysis, visualization development, dashboard, observations, and conclusion.

The dashboard screenshots and important observations can be included as part of the final documentation.

Proposed System Workflow

The proposed solution can be represented as the following workflow:

Smartphone Usage Data



Data Collection



Data Understanding



Data Cleaning & Preprocessing



Data Transformation



Data Analysis



Business Questions



Power BI Visualizations



Interactive Dashboard



Insights & Observations



Final Report

Main Features of the Proposed Solution

The proposed system will provide the following features:

Smartphone usage data analysis

1. Daily and weekend screen-time analysis
2. Age-wise usage analysis
3. Gender-wise comparison
4. Addiction-level analysis
5. Addicted vs. non-addicted analysis
6. Social media usage analysis
7. Gaming usage analysis
8. Sleep-related analysis
9. Notification analysis
10. Application-opening analysis
11. Stress-related analysis where available
12. KPI-based summary
13. Interactive charts and graphs
14. Filters and slicers
15. Relationship analysis
16. Key Influencers analysis where applicable
17. Interactive Power BI dashboard
18. Clear visual reporting
19. Easy exploration of smartphone usage patterns

2.3 Initial Project Planning

The initial project planning for Smartphone Addiction & Digital Well-Being Insights focuses on organizing the project activities before starting the actual development. The project is planned to analyze smartphone usage behavior and present meaningful insights using Microsoft Power BI.

The planning begins with understanding the problem and identifying the main objectives of the project. The required smartphone usage data is then identified and collected for analysis. The available data is examined to understand the different variables related to smartphone usage, such as screen time, social media usage, gaming, sleep, notifications, app usage, and addiction-related information.

After collecting the data, the project plans for data cleaning and preprocessing. The data is checked for missing values, duplicate records, inconsistent values, and incorrect data types. The cleaned data is then prepared for analysis in Power BI.

The next part of the planning is to identify the important questions that the project needs to answer. These questions focus on daily and weekend screen time, addiction levels, addicted and non-addicted users, age groups, gender, social media usage, gaming hours, sleep hours, notifications, and other available digital usage measures.

Based on these requirements, suitable Power BI visualizations are planned. Charts, graphs, KPI cards, filters, and interactive dashboard elements are selected to present the information clearly. Different visualizations are used to compare user groups, understand distributions, and observe relationships between selected variables.

The final stage of the initial planning is to develop an interactive Power BI dashboard containing the important findings of the analysis. The dashboard is planned to provide a clear and organized view of smartphone usage patterns and digital well-being indicators.

Overall, the initial project planning provides a structured path for the project, starting from problem identification and data collection, followed by data preparation, analysis, visualization, dashboard development, and final documentation. This planning helps ensure that the project activities are completed in an organized manner and that the final dashboard addresses the main objectives of the project.

3. Data Collection and Preprocessing Phase

The Data Collection and Preprocessing Phase is an important stage of the Smartphone Addiction & Digital Well-Being Insights project. In this phase, the required smartphone usage data is collected and prepared for further analysis. Since the accuracy of the final dashboard depends on the quality of the input data, the collected data must be properly examined, cleaned, and organized before using it for visualization.

The project uses smartphone usage information to study different aspects of users' digital behavior. The data includes information related to screen time, social media usage, gaming, sleep, notifications, app usage, age, gender, addiction levels, and other available digital well-being measures.

The first step is to identify and collect the required dataset. After collecting the data, it is examined to understand the available columns, values, categories, and data types. The dataset is then checked for common data-quality problems such as missing values, duplicate records, inconsistent values, and incorrect data types.

During preprocessing, unnecessary or incorrect data is handled, and the required values are organized into a suitable format. Numerical fields are prepared for calculations, while categorical fields are organized for comparisons and filtering. The prepared data is then used in Microsoft Power BI for creating charts, KPI cards, filters, and the interactive dashboard.

The main activities performed in this phase are:

- Data Collection: Obtaining the smartphone usage dataset required for the project.
- Data Understanding: Examining the structure and contents of the collected data.
- Data Cleaning: Identifying and handling missing, duplicate, or inconsistent records.
- Data Transformation: Preparing the available data into a suitable format for analysis.
- Data Type Checking: Ensuring that numerical, text, and categorical fields are represented appropriately.
- Data Organization: Arranging the required fields so they can be easily analyzed.
- Data Preparation for Power BI: Loading the prepared data into Power BI for further analysis and visualization.

The preprocessing phase ensures that the dataset is in a suitable condition for the next stages of the project. After completing this phase, the prepared data can be analyzed to identify smartphone usage patterns and create meaningful visualizations for the Smartphone Addiction & Digital Well-Being Insights dashboard.

3.1 Data Collection Plan and Raw Data Sources Identified

The Data Collection Plan and Raw Data Sources Identified stage focuses on identifying the data required for the Smartphone Addiction & Digital Well-Being Insights project. The main purpose of this stage is to collect relevant information that can be used to study smartphone usage behavior and analyze different aspects of digital well-being.

The project requires data related to users' smartphone activities and usage patterns. The planned data includes information such as age, gender, daily screen time, weekend screen time, social media usage, gaming hours, sleep hours, work/study hours, notifications received, app openings, stress level, addiction level, and addicted status, depending on the fields available in the project dataset.

Data Collection Plan

The data collection process is planned in the following manner:

1. Identify the required data – Determine the information needed to answer the project's analytical questions.

2. Identify the raw dataset – Select the smartphone usage dataset containing the required user and usage-related information.

3. Collect the raw data – Obtain the dataset in its original form before applying any cleaning or transformation.

4. Examine the dataset – Check the available columns, number of records, categories, numerical values, and data types.

5. Check data quality – Identify missing values, duplicate records, inconsistent entries, and other possible data-quality issues.

6. Prepare the data for preprocessing – Organize the collected raw data so that it can be cleaned and transformed before analysis.

7. Use the prepared data for analysis – After preprocessing, the data will be used in Power BI to create visualizations and the interactive dashboard.

Raw Data Sources Identified

The primary raw data source identified for the project is the Smartphone Usage Dataset. It provides information related to users' smartphone usage and digital behavior.

S.No.	Raw Data Source	Information Required	Purpose
1.	Smartphone Usage Dataset	Age, gender, screen time, social media, gaming, sleep, notifications, app usage and addiction-related information	To analyze smartphone usage patterns and digital well-being
2.	Project Dataset	User-level digital activity and related measurements available in the dataset	To create comparisons, calculations and Power BI visualizations

The collected raw data will serve as the foundation for the project. It will first be examined and preprocessed to improve its consistency and suitability for analysis. After preprocessing, the final dataset will be imported into Microsoft Power BI, where it will be used to create charts, KPI cards, filters, and the interactive dashboard.

Thus, the data collection plan ensures that the project has the required raw information for analyzing smartphone usage, screen time, digital activities, addiction-related patterns, and digital well-being.

3.2 Data Quality Report

The Data Quality Report describes the quality of the raw data collected for the Smartphone Addiction & Digital Well-Being Insights project. Before using the dataset for analysis and visualization, it is important to check whether the data is complete, consistent, accurate, and suitable for further processing.

The smartphone usage dataset contains information related to users' digital activities and digital well-being. The data is examined to identify possible issues such as missing values, duplicate records, incorrect data types, and inconsistent values.

These issues are considered during the preprocessing stage so that the data can be prepared properly for analysis in Power BI.

Data Source	Data Quality Issue	Severity	Resolution Plan
Smartphone Usage Dataset	Missing values in some fields	Moderate	Identify missing values and handle them appropriately according to the data and analysis requirement.
Smartphone Usage Dataset	Duplicate records	Low	Check for duplicate records and remove unnecessary duplicates.
Smartphone Usage Dataset	Incorrect data types	Moderate	Check the data types of all columns and convert them into appropriate formats.
Smartphone Usage Dataset	Inconsistent values	Moderate	Identify inconsistent entries and standardize them before analysis.
Smartphone Usage Dataset	Invalid or unexpected entries	Moderate	Examine unusual values and verify or correct them where appropriate.
Smartphone Usage Dataset	Different categories or formats	Low	Standardize categorical values and formats to ensure consistent analysis.

Data Quality Assessment

1. Completeness:

The dataset is checked for missing values in important fields. Missing information can affect calculations and visualizations, so such values need to be identified and handled during preprocessing.

2. Consistency:

The values in different columns are checked for consistency. Similar categories should follow a consistent format so that Power BI can group and analyze them correctly.

3. Accuracy:

The available values are examined for unusual or invalid entries. Values that do not follow the expected format or range are reviewed before being used for analysis.

4. Uniqueness:

The dataset is checked for duplicate records. Duplicate records can affect the total number of users and calculated measures, so unnecessary duplicates should be removed.

5. Validity:

The data types and formats of the columns are checked to ensure that the values are suitable for their intended purpose. Numerical fields should be usable for calculations, while categorical fields should be suitable for grouping and filtering.

6. Data Type Quality:

Columns containing numerical measurements such as screen time, gaming hours, social media hours, sleep hours, notifications, and app openings are checked to ensure that they can be used correctly for calculations and visualizations.

Resolution

The identified data-quality issues are handled during the preprocessing stage. Missing values are reviewed, duplicate records are removed where necessary, incorrect data types are converted, and inconsistent values are standardized. After these steps, the prepared dataset can be loaded into Power BI for further analysis.

Overall, the Data Quality Report helps ensure that the smartphone usage data is properly examined before creating the project's visualizations and dashboard.

3.3 Data Exploration and Preprocessing

The Data Exploration and Preprocessing stage is an important part of the Smartphone Addiction & Digital Well-Being Insights project. In this stage, the collected raw smartphone usage data is examined carefully and prepared for further analysis. The main purpose is to understand the structure and characteristics of the dataset and make sure that the data is suitable for creating accurate and meaningful visualizations in Power BI.

Data Exploration

Data exploration is the process of examining the collected dataset before performing detailed analysis. The dataset is studied to understand the available information, the number and type of variables, the values present in each field, and the relationships between different measurements.

For this project, the exploration focuses on smartphone usage and digital well-being information such as age, gender, daily screen time, weekend screen time, social media hours, gaming hours, sleep hours, work/study hours, notifications, app openings, stress level, addiction level, and addicted status, where these fields are available in the dataset.

During exploration, the data is examined to identify:

- The structure of the dataset.
- The different columns available.
- Numerical and categorical variables.
- The range and distribution of numerical values.
- Different categories present in categorical fields.
- Missing or empty values.
- Duplicate records.
- Inconsistent values.
- Incorrect data types.
- Unusual or unexpected entries.
- Data that may require transformation before visualization.

Exploring the data at this stage helps in understanding what information can be used to answer the project's business questions.

Data Preprocessing

After exploring the raw data, preprocessing is performed to prepare the dataset for analysis. Preprocessing involves cleaning, transforming, and organizing the data so that it can be used effectively in Power BI.

The major preprocessing activities include the following:

1. Handling Missing Values

The dataset is checked for empty or missing values. Missing values in important fields are identified and handled appropriately according to the requirements of the analysis.

2. Removing Duplicate Records

The data is checked for duplicate records. Unnecessary duplicate entries are removed so that they do not affect user counts, averages, or other calculations.

3. Checking Data Types

The data types of the columns are examined. Numerical fields should be available in a suitable numerical format for calculations, while categorical information should be stored appropriately for grouping and filtering.

4. Handling Inconsistent Values

The dataset is checked for inconsistent entries. Similar categories or values are standardized where necessary so that they are treated consistently during analysis.

5. Data Transformation

The required data may be transformed into a suitable structure for analysis. This can include organizing categories, preparing calculated fields, and arranging the data according to the requirements of the planned visualizations.

6. Preparing Data for Analysis

After cleaning and transformation, the dataset is reviewed again to ensure that the required fields are available and suitable for analysis. The prepared data is then used to create the Power BI visualizations.

7. Data Modelling

The prepared data is organized in Power BI so that the required calculations, relationships, filters, and visualizations can be created effectively. Measures and fields required for the dashboard are prepared according to the analysis requirements.

8. Saving the Processed Data

After completing the preprocessing activities, the prepared data is retained for further analysis and dashboard development.

Preprocessing Flow

Raw Smartphone Usage Data



Data Exploration



Check Missing Values



Check Duplicate Records



Check Data Types



Handle Inconsistent Values



Transform and Organize Data



Prepare Data for Analysis



Load into Power BI



Create Visualizations and Dashboard

Outcome of Data Exploration and Preprocessing

The outcome of this phase is a cleaned and organized dataset suitable for analysis and visualization. The prepared data can then be used to analyze screen-time behavior, addiction levels, age and gender differences, social media usage, gaming, sleep, notifications, app usage, and other available digital well-being measures.

This phase provides the foundation for the next stage of the project, where the prepared data is analyzed and converted into meaningful Power BI charts, KPI cards, filters, and an interactive dashboard.

4. Data Visualization

Data visualization is an important phase of the Smartphone Addiction & Digital Well-Being Insights project. After the data has been collected, explored, cleaned, and preprocessed, the prepared data is converted into meaningful visual representations using Microsoft Power BI. The main purpose of this phase is to present complex smartphone usage information in a simple, clear, and understandable form.

The project contains different types of information related to smartphone usage, including daily screen time, weekend screen time, social media usage, gaming hours, sleep hours, notifications, app openings, addiction levels, addicted status, age, and gender. Viewing all these values directly in a raw dataset can make it difficult to identify important patterns. Data visualization helps overcome this difficulty by representing the information through charts, graphs, KPI cards, and interactive dashboard elements.

In this project, different visualizations are selected according to the type of information that needs to be analyzed. Column charts and bar charts are used for comparing categories, while donut charts are used for showing proportions. A scatter chart is used to examine the relationship between screen time and sleep hours. KPI cards are used to display important numerical measures at a glance.

The visualization phase also includes the development of an interactive Power BI dashboard. The dashboard combines multiple visualizations into a single report page, allowing the user to view different aspects of smartphone usage together. Filters and slicers can be used to explore the available data according to selected categories such as gender, age group, addiction level, or other relevant fields.

The main visualizations developed for the project include:

- Average daily screen time by gender using a column chart.
- Distribution of addiction levels using a bar chart.
- Proportion of addicted and non-addicted users using a donut chart.
- Average screen time across age groups using a column chart.
- Relationship between daily screen time and sleep hours using a scatter chart.
- Comparison of gaming hours and social media hours using a column chart.
- Average notifications per day using a card.
- Overall digital usage measurements using KPI cards.

Through these visualizations, the project transforms the processed smartphone usage data into an easy-to-understand graphical format. The dashboard provides a centralized view of the available information and makes it easier to compare different categories, examine relationships, and identify patterns in smartphone usage and digital well-being.

Overall, the Data Visualization phase converts the results of data analysis into meaningful visual information and forms the main foundation of the final Power BI interactive dashboard.

4.1 Framing Business Questions

Framing business questions is an important step in the Smartphone Addiction & Digital Well-Being Insights project. After collecting, exploring, and preprocessing the smartphone usage data, clear questions are defined to identify what information needs to be analyzed from the dataset. These questions provide direction for the data analysis and help in selecting suitable Power BI visualizations.

The main purpose of framing business questions is to convert the general project objective into specific and measurable questions. Instead of simply looking at the raw data, the project focuses on finding meaningful information about users' smartphone usage, screen time, digital activities, addiction levels, and digital well-being indicators.

The business questions are framed based on the available variables in the smartphone usage dataset. These include age, gender, daily screen time, weekend screen time, social media hours, gaming hours, sleep hours, work/study hours, notifications, app openings, addiction level, and addicted status, where these fields are available in the project data.

The following business questions are considered for the analysis:

1. What is the average daily screen time by gender?

This question helps compare the average amount of daily smartphone screen time across different gender categories.

2. What are the different addiction levels among users?

This question focuses on understanding how users are distributed across the available addiction levels, such as None, Mild, Moderate, and Severe.

3. What is the proportion of addicted and non-addicted users?

This question helps understand the distribution of users according to their addicted status.

4. How does screen time vary across different age groups?

This question is used to examine differences in smartphone screen time among different age groups.

5. What is the relationship between screen time and sleep hours?

This question examines whether there is an observable relationship between the amount of time users spend on smartphones and their reported sleep hours.

6. How do gaming hours compare with social media hours?

This question compares two major digital activities and helps understand how users divide their smartphone usage between gaming and social media.

7. What is the average number of notifications received per day?

This question provides an overall measure of the notifications users receive and helps represent one aspect of their daily smartphone interaction.

8. What are the average digital usage patterns of users?

This question brings together important measures such as screen time, social media usage, gaming, sleep, notifications, app openings, and other available usage indicators.

These business questions help determine which calculations and visualizations need to be created in Power BI. For example, comparison-based questions can be represented using bar or column charts, proportion-based questions can be represented using a donut chart, relationship-based questions can be represented using a scatter chart, and important numerical values can be displayed using KPI cards.

Framing these questions also helps ensure that the dashboard remains focused on the main purpose of the project. Each visualization is created to answer a particular question rather than displaying data without a specific purpose. The questions therefore act as a connection between the prepared dataset, the analysis process, and the final interactive Power BI dashboard.

Overall, the business questions provide a structured foundation for analyzing smartphone usage behavior and digital well-being. They help identify the required measures, comparisons, relationships, and patterns and guide the development of meaningful visualizations for the final dashboard.

4.2 Developing Visualizations

Developing visualizations is an important stage of the Smartphone Addiction & Digital Well-Being Insights project. After framing the business questions, the next step is to convert the prepared data into meaningful charts, graphs, and summary indicators using Microsoft Power BI. The main purpose of developing visualizations is to present smartphone usage information in a simple, clear, and understandable form so that important patterns and comparisons can be identified easily.

The prepared smartphone usage dataset contains different types of information related to users' digital activities. These include daily screen time, weekend screen time, social media hours, gaming hours, sleep hours, work/study hours, notifications, app openings, age, gender, addiction level, and addicted status, depending on the available fields in the project data. Since these values are difficult to understand when viewed only as raw records, appropriate visualizations are developed to represent the information graphically.

The visualization development process begins by selecting the appropriate fields and measures from the prepared dataset. Each business question is matched with a

suitable type of visualization. The selection of the visualization depends on the type of information that needs to be represented. For example, comparisons between categories can be displayed using bar or column charts, proportions can be displayed using a donut chart, relationships between numerical variables can be displayed using a scatter chart, and important numerical measures can be displayed using KPI cards.

For the project, the following visualizations are developed:

- **Average Daily Screen Time by Gender:** A column chart is used to compare the average daily screen time among different gender categories. This makes it easier to observe differences in screen-time usage between groups.
- **Addiction Levels Among Users:** A bar chart is used to display the number of users across different addiction levels. This provides a clear view of how users are distributed among the available addiction categories.
- **Addicted and Non-Addicted Users:** A donut chart is used to represent the proportion of users based on their addicted status. The visualization provides a simple view of the distribution between the two categories.
- **Screen Time by Age Group:** A column chart is used to compare screen-time patterns across different age groups. This helps identify differences in smartphone usage among age categories.
- **Screen Time and Sleep Relationship:** A scatter chart is developed to examine the relationship between daily screen time and sleep hours. Each user's values can be represented as a data point, allowing the overall pattern between the two measures to be observed.
- **Gaming Hours and Social Media Hours:** A column chart is used to compare average gaming hours and social media hours. This helps understand the contribution of these activities to users' digital usage.
- **Average Notifications:** A card visual is used to display the average number of notifications received per day. This provides a quick summary of notification activity.
- **Overall Digital Usage KPIs:** KPI cards are used to display important measures such as average daily screen time, average social media hours, average gaming hours, average sleep hours, average notifications, average app openings, and other relevant measures available in the dataset.

The visualizations are then arranged together to form an interactive Power BI dashboard. Filters and slicers can be used to allow the user to explore the data based on categories such as age, gender, addiction level, or other available fields. When a filter or selection is applied, the relevant visualizations can update accordingly, making the dashboard useful for interactive analysis.

The development of visualizations also involves checking whether each chart clearly answers its corresponding business question. Titles, labels, measures, and categories are arranged so that the information can be understood without examining the raw dataset. The visualizations are organized in a logical manner so that the dashboard provides a clear overall view of smartphone usage behavior and digital well-being indicators.

Overall, the visualization development stage converts the processed smartphone usage data into meaningful graphical information. By using appropriate Power BI charts, KPI cards, filters, and interactive dashboard elements, the project makes it easier to analyze screen-time behavior, digital activities, addiction levels, sleep patterns, and other digital well-being indicators. These visualizations form the main analytical component of the final Power BI dashboard.

5. Dashboard

The Dashboard is the main output of the Smartphone Addiction & Digital Well-Being Insights project. It brings together the important analysis results and visualizations developed during the previous stages into a single interactive Power BI report. The purpose of the dashboard is to provide a clear and organized view of smartphone usage behavior and help users understand different digital activity patterns.

The dashboard is designed using Microsoft Power BI after completing the data collection, data preprocessing, business-question framing, and visualization development stages. The prepared smartphone usage data is connected to Power BI, and the required measures and fields are used to create different charts, KPI cards, and interactive elements.

The dashboard focuses on important aspects of smartphone usage, including daily screen time, weekend screen time, social media usage, gaming hours, sleep hours, work/study hours, notifications, app openings, age, gender, addiction levels, and addicted status, based on the available project data. These measures are presented through different visual elements so that the information can be understood more easily than through raw data.

The dashboard contains KPI cards that provide quick information about important numerical measures. These cards can be used to display values such as average daily screen time, average social media hours, average gaming hours, average sleep hours, average notifications, average app openings, average weekend screen time, average work/study hours, total users, and average age.

Different charts are used to answer the business questions identified in the project. A column chart can be used to compare average daily screen time by gender and screen time across age groups. A bar chart can represent the distribution of users across different addiction levels. A donut chart can show the proportion of addicted and non-addicted users. A scatter chart can represent the relationship between

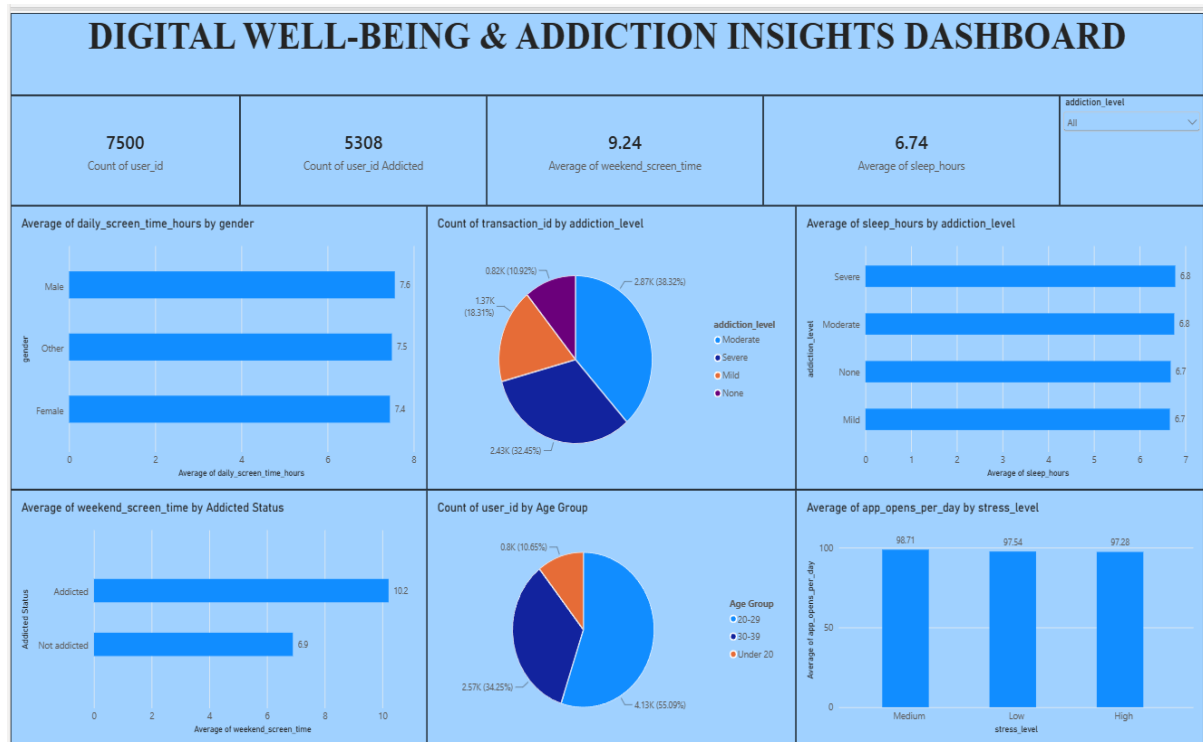
screen time and sleep hours. Another comparison chart can be used to compare gaming and social media usage.

The dashboard also includes interactive filtering features. Slicers and filters allow the data to be examined according to available categories such as gender, age group, addiction level, or addicted status. When a user selects a particular category, the related visualizations can change according to the selected filter. This makes the dashboard more useful for exploring different user groups.

The dashboard provides a centralized view of the project analysis. Instead of checking each individual calculation separately, users can observe several important measures and visualizations on one report page. This makes it easier to identify usage patterns, compare categories, and examine relationships between different digital activities.

The dashboard is also designed to maintain a clear connection between the business questions and visualizations. Each major visual is included for a specific analytical purpose. For example, screen-time charts address screen-time comparisons, addiction-level charts address addiction distribution, and the screen-time versus sleep visualization addresses the relationship between these two measures.

Overall, the Power BI dashboard transforms the prepared smartphone usage data into an interactive and understandable analytical report. It provides a single platform for examining smartphone usage patterns, digital activities, addiction-related information, and digital well-being indicators. The completed dashboard represents the final visualization stage of the project and provides a clear presentation of the analysis results.



6. Report

The Report is the final stage of the Smartphone Addiction & Digital Well-Being Insights project. It presents the overall analysis and findings obtained from the smartphone usage dataset in a structured and understandable manner. The report documents the complete process followed in the project, starting from problem identification and data collection and continuing through data preprocessing, business-question framing, visualization development, and dashboard creation.

The main purpose of the report is to provide a clear summary of the information analyzed through Microsoft Power BI. It explains how smartphone usage data was transformed into meaningful information through different charts, graphs, KPI cards, and interactive dashboard elements. The report also helps document the important observations obtained from the analysis.

The report covers different aspects of smartphone usage behavior, including daily screen time, weekend screen time, social media usage, gaming hours, sleep hours, work/study hours, notifications, app openings, age, gender, addiction levels, and addicted status, based on the fields available in the project dataset. These factors are considered to understand different patterns of digital usage and digital well-being.

The report explains the business questions considered during the analysis. These questions focus on comparing screen time by gender and age group, understanding addiction-level distribution, examining addicted and non-addicted users, comparing

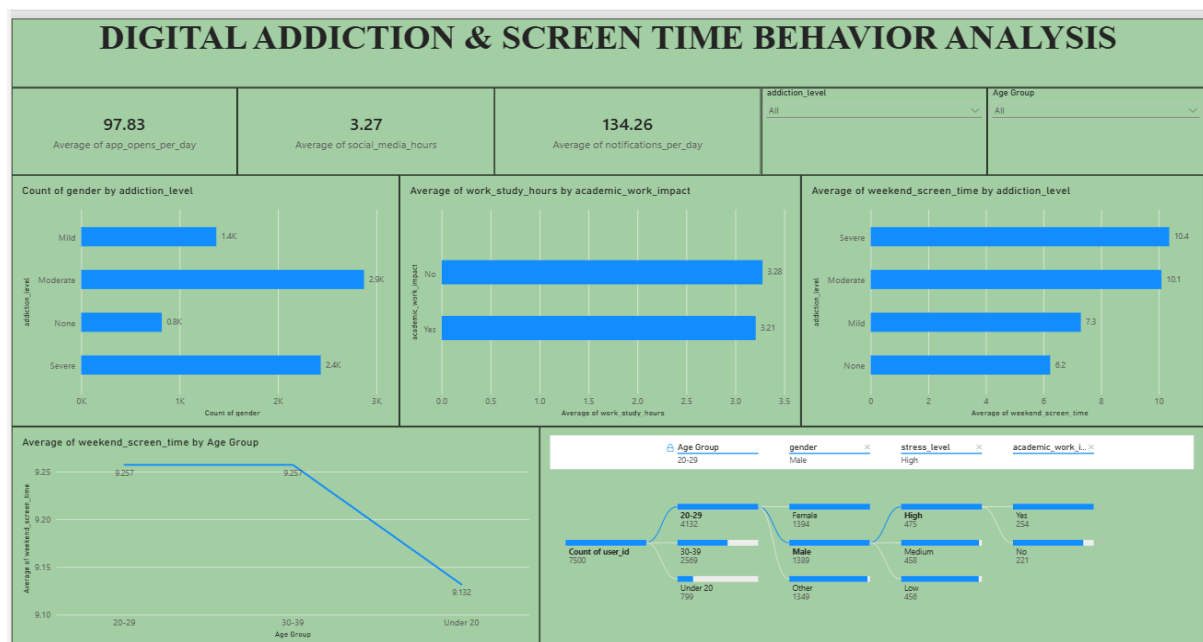
gaming and social media usage, studying the relationship between screen time and sleep, and identifying overall digital usage patterns.

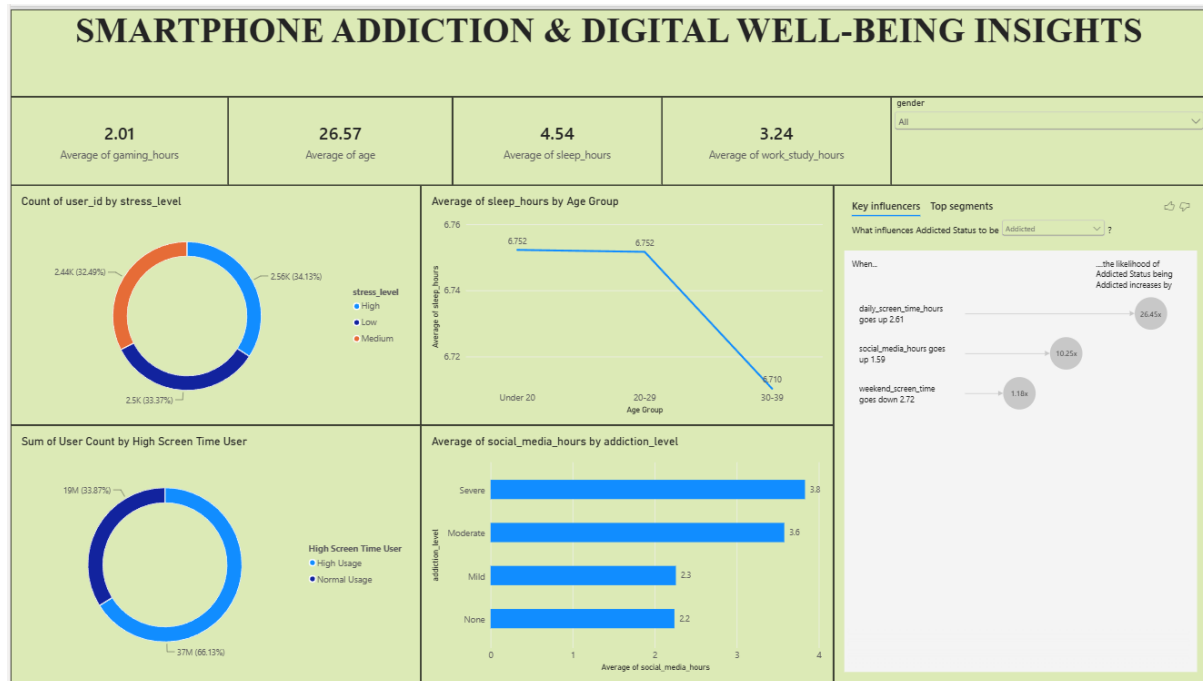
The report also documents the visualizations developed to answer these questions. Column charts, bar charts, donut charts, scatter charts, and KPI cards are used to present the analyzed information. Each visualization provides a specific view of the data and supports the interpretation of the corresponding business question.

Another important part of the report is the presentation of the Power BI dashboard. The dashboard brings the major visualizations and important measures together in one place. Interactive filters and slicers allow the information to be explored according to the available categories. This provides a convenient way to examine different groups and understand their smartphone usage patterns.

The report also describes the overall data-analysis process followed in the project. The raw data was first examined and prepared, followed by data cleaning and preprocessing. After that, business questions were framed, suitable visualizations were developed, and the visualizations were arranged into the Power BI dashboard. The final report documents these stages and explains how they contribute to the overall project objective.

The final report therefore acts as a complete record of the Smartphone Addiction & Digital Well-Being Insights project. It provides a structured explanation of the problem, proposed solution, data preparation, business questions, visualizations, dashboard, and analysis. It helps the reader understand how the project uses Power BI to convert smartphone usage data into meaningful and easily understandable digital well-being insights.





7. Performance Testing

Performance testing is an important stage of the Smartphone Addiction & Digital Well-Being Insights project. It is carried out to check whether the Power BI dashboard and its visualizations work properly and provide information efficiently. After developing the dashboard and report, performance testing helps ensure that the completed project is usable, responsive, and capable of handling the prepared smartphone usage data.

The main purpose of performance testing is to examine how the dashboard behaves when different visualizations, filters, slicers, and data interactions are used. Since the project contains multiple charts and numerical measures, it is important to verify that the dashboard responds correctly when users interact with different components.

The performance testing process begins by checking whether the prepared dataset loads correctly into Power BI. The data should be available without unnecessary delays, and the required fields should be accessible for creating calculations and visualizations. The dashboard is then tested to ensure that all major visual elements display the expected information.

The KPI cards are checked to confirm that the required measures are displayed correctly. Measures such as average daily screen time, average social media hours, average gaming hours, average sleep hours, average notifications, average app openings, weekend screen time, work/study hours, total users, and average age are checked against the available data.

The different charts are also tested individually. The column charts are checked to ensure that category comparisons are displayed correctly. The bar chart used for addiction levels is checked to ensure that the available addiction categories are represented correctly. The donut chart for addicted and non-addicted users is checked to ensure that the categories and proportions are displayed properly.

The scatter chart used to examine the relationship between screen time and sleep hours is also tested. The values represented in the chart should correspond to the selected fields from the dataset. The visualization should respond correctly when filters or selections are applied.

Another important part of performance testing is checking the interactive features of the dashboard. Slicers and filters are tested by selecting different categories such as age group, gender, addiction level, or addicted status. After a selection is made, the related charts and KPI cards are checked to ensure that they respond appropriately to the selected data.

The dashboard is also tested for visual consistency and readability. Chart titles, labels, values, legends, and other dashboard elements are checked to ensure that they are clearly visible. The arrangement of the visualizations is reviewed so that charts do not overlap and important information is not hidden.

Performance testing also includes checking the correctness of the calculations used in the dashboard. Aggregations such as average, count, and other measures are reviewed to make sure that they represent the intended information. Incorrect field selection, inappropriate aggregation, or calculation errors can affect the final analysis, so these aspects are checked during testing.

The report and dashboard are tested under different user interactions. For example, the dashboard can be viewed without applying any filter and then tested again after selecting individual categories. This helps determine whether the interactions between different visuals are working correctly.

The following areas are considered during performance testing:

- Data loading: Check whether the prepared smartphone usage data loads correctly.
- Data accessibility: Verify that required fields and measures are available for analysis.
- Visualization loading: Check whether charts and KPI cards display properly.
- Calculation performance: Verify that calculated measures provide the expected results.
- Filter performance: Check whether slicers and filters respond correctly.
- Interactive performance: Verify that related visualizations update after selections.

- Dashboard responsiveness: Check whether the dashboard remains usable during interactions.
- Visual readability: Check titles, labels, values, legends, and chart arrangements.
- Data consistency: Ensure that different visualizations represent the selected data consistently.
- Report performance: Check whether the completed report provides information in an organized and usable manner.

Performance testing can be carried out using a simple testing process:

Prepared Data → Load into Power BI → Check Data → Test Measures → Test Visualizations → Test Filters/Slicers → Test Dashboard Interactions → Check Readability → Verify Results → Final Dashboard

The testing process helps identify possible problems before the final dashboard is presented. If a visualization does not respond correctly, a measure displays an unexpected result, or a filter does not affect the required visual, the corresponding Power BI configuration can be reviewed and corrected.

Performance testing also helps ensure that the dashboard remains focused on its intended purpose. The project contains several visualizations related to smartphone usage and digital well-being, so each visual should provide useful information without creating unnecessary complexity. The dashboard should allow users to move between different views of the data and understand the results without difficulty.

The final stage of testing is to review the complete dashboard as a single report. All KPI cards, charts, filters, slicers, and other available interactive elements are checked together. The final dashboard is then reviewed to ensure that the visualizations work consistently and that the information presented is suitable for answering the framed business questions.

Overall, performance testing ensures that the Smartphone Addiction & Digital Well-Being Insights dashboard functions correctly after development. It checks the data loading, calculations, visualizations, interactions, filters, and overall dashboard usability. This testing provides greater confidence that the final Power BI report can be used to analyze smartphone usage patterns and present digital well-being information in a clear and organized manner.

7.1 Utilization of Data Filters

Data filters are an important feature used in the Smartphone Addiction & Digital Well-Being Insights Power BI dashboard. They allow users to focus on a particular part of the dataset instead of viewing all records at the same time. By applying filters, the dashboard can provide a more specific view of smartphone usage behavior and make the analysis easier to understand.

In this project, the smartphone usage dataset contains information related to different users and their digital activities. The dashboard contains measures and visualizations related to daily screen time, weekend screen time, social media hours, gaming hours, sleep hours, work/study hours, notifications, app openings, age, gender, addiction level, and addicted status, according to the available project data. Data filters can be used to examine these measures for selected groups of users.

The main purpose of using filters is to make the dashboard interactive and user-friendly. Without filters, the visualizations show an overall view of the available data. When a filter is applied, the charts and other related dashboard elements can be examined for the selected category. This allows users to perform more focused analysis.

For example, a Gender filter can be used to select a particular gender category. After making the selection, the relevant visualizations can show the smartphone usage information for that selected group. This can help in examining differences in daily screen time, social media usage, gaming, sleep, and other available measures.

An Age or Age Group filter can be used to focus the analysis on a particular age category. This allows the user to examine screen-time behavior and other digital activities within the selected age group rather than considering the complete dataset.

An Addiction Level filter can be used to focus on the available addiction categories. By selecting a particular level, the dashboard can be used to examine the related usage patterns of users belonging to that category. Similarly, an Addicted Status filter can be used to view information for the available addicted or non-addicted categories.

Filters can also be used together. For example, a user may select an age group and then select an addiction level. The dashboard can then be examined according to the selected combination, provided those fields and interactions are configured in the report. This makes it possible to perform more detailed analysis of specific groups.

The utilization of data filters can be explained through the following process:

Complete Dataset → Select Filter → Choose Category → Dashboard Updates → Examine Selected Data → Compare Results → Identify Patterns

Data filters are useful for several purposes in this project:

- Reducing the amount of data displayed: Filters allow users to focus only on the records relevant to their analysis.
- Comparing user groups: Different categories can be selected to examine differences in smartphone usage.
- Analyzing age groups: Users can focus on particular age categories and examine their usage patterns.
- Analyzing gender categories: Screen-time and other digital activities can be examined for selected gender categories.

- Examining addiction levels: Users can focus on particular addiction-level categories.
- Examining addicted status: The dashboard can be filtered according to the available addicted-status categories.
- Studying digital activities: social media, gaming, sleep, notifications, and app-opening measures can be examined for selected groups.
- Improving dashboard interaction: Filters allow users to explore the report according to their requirements.
- Supporting business questions: Filters help users obtain more focused information from the visualizations.

The filters also improve the usability of the dashboard because users do not need to manually examine individual records in the raw dataset. Instead, they can select a required category and use the Power BI visuals to observe the corresponding information. This makes the analysis faster and easier to interpret.

When filters are applied, it is also important to check that the dashboard visuals respond correctly. KPI cards, charts, and other interactive elements should reflect the selected filter where the Power BI interaction is configured to do so. Testing these interactions helps ensure that the dashboard provides consistent information.

For example, when a particular age group is selected, the user can observe the corresponding changes in the screen-time visualization and other related measures. When an addiction category is selected, the user can examine the digital usage characteristics represented by the available visuals. These filtered views provide a more detailed way of exploring the dataset.

Overall, the utilization of data filters makes the Smartphone Addiction & Digital Well-Being Insights dashboard more interactive and flexible. Filters help users move from an overall view of smartphone usage to a focused view of selected user groups. They support comparison, exploration, and interpretation of the available data and make the Power BI dashboard more useful for understanding smartphone usage behavior and digital well-being indicators.

7.2 Number of Calculation Columns / Measures

Calculation columns and measures are important components used in the Smartphone Addiction & Digital Well-Being Insights Power BI project. They help convert the available raw data into useful calculated values that can be displayed in the dashboard. Calculations make it easier to summarize smartphone usage information and provide meaningful values for the different business questions.

In Power BI, a calculated column is created when a new value needs to be calculated for each row of the dataset. A measure is a calculation that is evaluated based on the data currently being viewed, including the filters and selections applied

by the user. Measures are particularly useful for creating dynamic summaries in an interactive dashboard.

For this project, calculations are used to analyze important smartphone usage indicators such as daily screen time, social media hours, gaming hours, sleep hours, weekend screen time, work/study hours, notifications, app openings, age, and user count, according to the available dataset fields.

The main measures required for the dashboard include:

- 1. Total Users** – Used to calculate the total number of users available in the dataset.
- 2. Average Age** – Calculates the average age of the users.
- 3. Average Daily Screen Time** – Calculates the average number of hours users spend on their smartphones each day.
- 4. Average Weekend Screen Time** – Calculates the average smartphone screen time during weekends.
- 5. Average social media Hours** – Calculates the average amount of time users spend on social media.
- 6. Average Gaming Hours** – Calculates the average amount of time users spend on gaming.
- 7. Average Sleep Hours** – Calculates the average number of sleep hours represented in the dataset.
- 8. Average Work/Study Hours** – Calculates the average amount of time spent on work or study.
- 9. Average Notifications** – Calculates the average number of notifications received by users.
- 10. Average App Openings** – Calculates the average number of times users open applications.

These measures support the KPI cards and other visualizations used in the dashboard. For example, the Average Daily Screen Time measure can be used in a KPI card or in a column chart to compare screen time across gender or age groups. Similarly, average gaming and social media measures can be used to compare different types of digital activities.

Calculated columns can also be useful when the raw data needs to be organized into meaningful categories. For example, an Age Group column can categorize users into groups based on age, if such categorization is required by the project. These categories can then be used in charts and filters.

The calculation measures also work with the dashboard's filters. When a user selects a particular age group, gender, addiction level, or other available category, the relevant measures can recalculate according to the selected data. This makes the dashboard dynamic and allows users to examine specific groups without manually changing the underlying dataset.

The calculation process can be represented as:

Raw Dataset → Select Required Field → Create Calculation → Validate Result → Use in Visualization → Display in Dashboard

The number of calculation columns and measures should be documented according to the actual Power BI model created for the project. The uploaded PDF could not be extracted as readable text, so the exact number of calculated columns/measures created in the original Power BI file cannot be verified from that PDF alone. The list above represents the calculations relevant to the dashboard analysis described in this documentation, rather than a verified count of the actual .pbix model.

Overall, calculation columns and measures provide the required numerical foundation for the Power BI dashboard. They help summarize the smartphone usage data, support the business questions, power KPI cards and charts, and allow the dashboard to provide dynamic results when filters are applied.

7.3 Number of Visualizations

The Number of Visualizations represents the different charts, graphs, KPI cards, and other visual elements developed in the Smartphone Addiction & Digital Well-Being Insights Power BI dashboard. Visualizations are used to convert the prepared smartphone usage data into a graphical form so that users can understand the information more easily.

The project uses different types of visualizations because each business question requires a suitable method of presentation. A single type of chart cannot effectively represent all the information in the dataset. Therefore, different visualizations are selected according to whether the requirement is to compare categories, show proportions, identify relationships, or display important numerical values.

The major visualizations developed for the project include the following:

1. Average Daily Screen Time by Gender – Column Chart

This visualization compares the average daily screen time for the available gender categories. It helps present screen-time differences between groups in an easy-to-understand graphical format.

2. Addiction Levels – Bar Chart

This visualization represents the distribution of users across the available addiction levels. It provides a clear comparison of the number of users in each addiction category.

3. Addicted vs Non-Addicted Users – Donut Chart

This visualization represents the proportion of users according to their addicted status. It provides a simple part-to-whole view of the available categories.

4. Screen Time by Age Group – Column Chart

This visualization compares smartphone screen-time patterns across different age groups. It helps examine how usage varies between the available age categories.

5. Screen Time vs Sleep Hours – Scatter Chart

This visualization represents the relationship between daily screen time and sleep hours. Each observation can be represented as a point, allowing the overall relationship between the two numerical measures to be examined.

6. Gaming Hours vs Social Media Hours – Column Chart

This visualization compares the average time spent on gaming and social media activities. It provides a graphical representation of these two major digital activities.

7. Average Notifications – Card Visual

A card visual is used to display the average number of notifications received by users. It provides an immediate summary of notification activity.

8. Digital Usage KPI Cards

KPI cards are used to display important summary measures from the dataset. These can include average daily screen time, average social media hours, average gaming hours, average sleep hours, average weekend screen time, average work/study hours, average notifications, average app openings, total users, and average age, according to the dashboard configuration.

The visualizations are arranged together within the Power BI dashboard to provide a centralized view of the analysis. Charts are used for comparisons and relationships, while cards are used for important numerical summaries. Filters and slicers can also be connected to these visualizations so that users can explore the data based on selected categories.

The number of visualizations should be counted from the actual final Power BI dashboard, because the uploaded PDF available in the conversation could not be read as machine-readable text and therefore does not provide a verifiable count of the visuals. The list above documents the visualizations described for the project; it should not be treated as a verified numerical count of the .pbix file.

Overall, the visualizations form the core of the Power BI dashboard. They transform smartphone usage data into graphical information and help users understand screen time, addiction levels, digital activities, sleep patterns, and other digital well-being indicators in a clear and organized manner.

8. Conclusion / Observation

The Smartphone Addiction & Digital Well-Being Insights project was developed to analyze smartphone usage behavior and present the available information in a clear and meaningful way using Microsoft Power BI. The project converts smartphone usage data into useful visual information through data preprocessing, business-question framing, calculations, visualizations, and an interactive dashboard.

The analysis focuses on different aspects of users' digital activities, including daily screen time, weekend screen time, social media usage, gaming hours, sleep hours, work/study hours, notifications, app openings, age, gender, addiction levels, and addicted status, based on the available fields in the project dataset. By bringing these measures together, the dashboard provides an organized view of different smartphone usage patterns.

One of the important observations from the project is that smartphone usage can be examined from multiple perspectives rather than using only one measure. Daily screen time and weekend screen time provide information about overall phone usage, while social media and gaming hours provide information about specific digital activities. Similarly, notifications and app openings provide additional indicators of how frequently users interact with their smartphones.

The project also provides an analysis of addiction levels and addicted status. The addiction-level visualization helps organize users according to the available categories, while the addicted-status visualization provides a broader view of the available user groups. These visualizations make it easier to examine the distribution of users within the dataset.

Another observation is that age and gender can be used as important categories for exploring differences in smartphone usage. By applying appropriate filters and visualizations, the dashboard allows users to examine screen-time and other available digital activity measures for selected groups.

The relationship between screen time and sleep hours is also included as an important area of observation. A scatter visualization allows the values of these two measures to be viewed together. This provides a way to examine the pattern present in the dataset without relying only on individual values.

The comparison between gaming hours and social media hours provides another perspective on digital activity. Instead of considering smartphone usage as a single measure, the project separates different activities and allows them to be compared through visualizations.

The KPI cards provide a quick summary of the major numerical measures. Based on the dashboard values shown in the project screenshots, the dashboard presents measures such as average daily screen time, average social media hours, average gaming hours, average sleep hours, average notifications, average app openings, average weekend screen time, average work/study hours, total users, and average age. These values provide an overall summary of the dataset represented in the dashboard.

The use of Power BI filters and slicers makes the analysis interactive. Users can select available categories and examine the corresponding information in the dashboard. This reduces the need to manually search through raw records and makes it easier to focus on specific groups or usage patterns.

The project also demonstrates the importance of data preprocessing and visualization in data analysis. Raw data by itself may be difficult to interpret when it contains many records and variables. After the data is prepared and represented through suitable visualizations, the information becomes easier to understand and compare.

The observations obtained from the project should be interpreted as patterns within the available dataset. The dashboard shows relationships and distributions in the data, but these observations alone should not be treated as proof that one factor directly causes another. For example, a visualization showing screen time and sleep together can show an observed pattern in the dataset, but it does not by itself establish causation.

Overall, the project successfully provides a structured approach to analyzing smartphone usage and digital well-being indicators. The combination of data preparation, calculations, business questions, Power BI visualizations, filters, and dashboard presentation makes the analysis organized and accessible. The completed dashboard can be used to explore the available smartphone usage data and obtain a clear overview of screen-time behavior, digital activities, addiction-related categories, and related digital well-being measures.

9. Future Scope

The Smartphone Addiction & Digital Well-Being Insights project can be further improved and expanded in the future by adding more data, advanced analysis techniques, and additional interactive features. The current project focuses on analyzing the available smartphone usage data through Microsoft Power BI. Future development can extend this analysis to provide a broader and more detailed understanding of digital usage behavior.

One important area of future development is the collection of larger and more diverse datasets. The project can be extended by collecting smartphone usage

information from a larger number of users and from different groups. A larger dataset can provide more information for comparison and can make it possible to examine a wider range of usage patterns.

The project can also be extended by including additional smartphone usage parameters. More information about application categories, communication activities, entertainment usage, study-related usage, work-related usage, and other digital activities can be included when such data is available. This would allow the dashboard to provide a more detailed view of how users spend their smartphone time.

Another possible improvement is time-based analysis. Instead of analyzing only summarized usage values, future versions can include data collected over different days, weeks, or months. This would allow changes in screen time and other digital activities to be observed over time and could provide a better understanding of changing usage patterns.

The Power BI dashboard can also be enhanced with more advanced interactive features. Additional slicers, drill-through pages, tooltips, bookmarks, and navigation options can be introduced to make the dashboard easier to explore. Users could move from an overall summary to more detailed views of specific categories or groups.

Future versions can also include advanced analytical techniques. Statistical analysis or machine-learning methods could be explored to identify patterns within the available smartphone usage data. Such techniques could be used to study relationships between different variables and identify groups with similar usage patterns. Any such analysis would need to be validated using appropriate data and methods.

Another area of future scope is the development of personalized digital well-being insights. If suitable user-level data is available and handled appropriately, the system could provide individualized summaries of screen time, social media usage, gaming, sleep, notifications, and other usage indicators. This could make the analysis more relevant to individual usage patterns.

The project can also be expanded by adding automated data updates. Instead of manually loading updated datasets, future implementations could connect Power BI to a suitable regularly updated data source. This would allow the dashboard to display more recent information when the data source supports such updates.

The visualization section can also be improved by adding more analytical visuals and detailed report pages. Different charts can be introduced to examine additional relationships between smartphone usage variables. Separate pages could be created for screen-time analysis, addiction-level analysis, digital activities, sleep-related analysis, and overall digital well-being, depending on the available data.

Future development can also focus on improving the user experience of the dashboard. Clearer navigation, meaningful titles, informative tooltips, consistent formatting, and well-organized report pages can make the dashboard easier for different users to understand and use.

The project may also be extended to compare different time periods or user groups when suitable data becomes available. Such comparisons could help identify changes in usage patterns and differences between selected groups without relying only on an overall average.

Data quality can also be improved in future versions by establishing a more systematic process for checking missing values, duplicate records, inconsistent categories, incorrect data types, and unusual entries before the data is loaded into Power BI. Better data quality can improve the reliability and consistency of the resulting visualizations.

Overall, the future scope of the project includes larger datasets, additional usage variables, time-based analysis, advanced Power BI features, improved visualizations, automated data updates, advanced analytical methods, and more detailed digital well-being analysis. These improvements can make the project more comprehensive while continuing to use data analysis and visualization to understand the smartphone usage patterns represented in the available data.

10.APPENDIX

GitHub link:

<https://github.com/easha2005/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics>

Project demo link:

<https://github.com/easha2005/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics/blob/main/7.PROJECT%20DOCUMENTATION/project%20demonstration.mp4>